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ÇALIK ENERJİ SAN. VE TİC. A.Ş

CAP DES BICHES COMBINED CYCLE POWER STATION PROJECT (300 MW)

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ELECTRICAL DESIGN CRITERIA

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1.0 PREAMBLE

West African Energy (WEA) intends to build a combined cycle power plant at Dakar. ÇALIK ENERJİ (CE) has received the order from WEA to execute the power plant on an Engineering, Procurement, Construction and Commissioning (EPC) basis. CE has appointed Fichtner India (FI) as Engineering Sub Contractor for the power plant. FI has brought out Design criteria for the Electrical system for the power plant, in this document.

2.0 PROJECT DESCRIPTION

The project is located at Cap des Biches site, on the edge of the Atlantic Ocean, located in the North Industrial Zone of Dakar, 25 KM from the port of Dakar.

- Altitude : 20 m + NGA approximately.
- Ambient temperature : Design: 26° C, Max: 41° C , Min: 10 °C
- Relative humidity : Design: 70%, Max: 98% , Min: 32%
- Seismicity : ZONE 0 in accordance with UBC 97
- Wind direction : North and North-West
- Ambient Air : Salty air environment with dust concentration
- Basic wind speed : 161km / hour
- Annual Rain fall : 2500 mm

The plant configuration comprise of One (1) combined cycle power block in multi shaft configuration. The block consists of two (2) gas turbine and generators (GTGs), two (2) heat recovery steam generators (HRSGs) & one (1) steam turbine and generator (STG) and Balance of Plant (BOP) equipment. The gas turbine will be of Frame 9E.03. The Steam turbine will be of axial exhaust type with air cooled condenser. Power generated from GTG s and STG will be stepped up and evacuated to 225 kV grids and 90 kV grids.

3.0 LIST OF ABBREVIATIONS

The following abbreviations shall be used in the Design Basis Report:

EHV	Extra High Voltage
HV	High Voltage
MV	Medium Voltage
LV	Low Voltage
AC	Alternating Current
DC	Direct Current
MCC	Motor Control Center
PMCC	Power Motor Control Center
THD	Total Harmonic Distortion
UPS	Uninterruptible Power Supply system
IP	Mechanical Degree of Protection
OH	Overhead
AVR	Automatic Voltage Regulator
ECP	Electrical Control Panel
UCP	Unit Control Panel
CT	Current Transformer
VT	Voltage Transformer

SF6	Sulphur Hexafluoride Gas
S/S	Substation
PVC	Polyvinyl Chloride
XLPE	Cross-linked polyethylene
NC	Normally Closed
NO	Normally Open
GIS	Gas Insulated Switchgear
ASB	Auxiliary Services Board
ACSR/AW	Aluminium Conductor Aluminium Clad Steel Reinforced
OFGW	Optical Fiber Ground Wire
MMS	Machine Monitoring System
DCS	Distributed Control System
OGW	Overhead Ground Wire
EPC	Engineering, Procurement, Construction and Commissioning
GTG	Gas Turbine and Generator
HRSG	Heat Recovery Steam Generator
STG	Steam Turbine and Generator
BOP	Balance of Plant
UAT	Unit auxiliary transformer
GSUT	Generator Step-up transformer
BSDG	Black start Diesel Generator
GCB	Generator Circuit Breaker
IPDB	Isolated Phase Bus Duct
TEWAC	Totally Enclosed with Water to Air Heat Exchanger
UPS	Un Interrupted Power supply
VFD	Variable Frequency Drives

4.0 PLANT ELECTRICAL SYSTEM

Power will be generated at 15 kV, 50 Hz by the two gas turbine generators and one steam turbine generator. The generators are connected to Step-up transformers for power evacuation, through isolated phase duct and generator circuit breaker.

The bus ducts of the gas turbine generators include a tap connection for Unit auxiliary transformer (UAT) to supply the medium voltage (MV) switchgears for auxiliary power.

Gas turbine Generator step up transformers are connected to the 225kV and 90kV GIS Switchyard with EHV cable connection and ST Generator step up transformer is connected to 225kV GIS with EHV cable connection.

The gas turbine uses a Cranking Motor supplied from the 6.6kV MV switchgear. The Cranking Motor starts turning the generator and accelerates it from turning gear speed to firing speed. Auto Transformer is envisaged for starting the Cranking motor. Once the unit is fired and self-sustaining, the Cranking Motor is switched off.

The GIS switchyard breaker is kept closed under normal operating conditions. During start up, power for the plant's auxiliaries and Cranking Motor system is provided by the unit auxiliary transformer from the 225 or 90 kV external grids and through the Generator step up transformer while the MV generator circuit breaker is open. After the generator reaches proper voltage and frequency, it is synchronized to the system by closing its MV generator circuit breaker.

In case of the steam turbine generator, the synchronization takes place by means of closing the 225 kV GIS Switchyard circuit breaker (or) MV generator circuit breaker (GCB) after the Steam turbine generator reaches proper voltage and frequency.

The Electrical system comprises the following sub systems:

- Power Generation
- Power Evacuation
- Auxiliary power supply
- Black start DG system
- Emergency AC supply
- 125 V DC supply
- UPS Supply

The plant comprises the following major electrical equipment:

- Generators
- Generator step up Transformers
- 225 kV & 90 kV GIS Substation
- Unit Auxiliary Transformers
- Service Transformers
- Bus duct
- Generator Circuit Breaker
- Control & Relay panels
- MV Switchgear
- LV Switchgear
- Battery & Charger
- UPS equipment
- Black start diesel Generator set
- Power & Control cables
- Motors
- Cabling, Earthing & Lightning system
- Illumination system

5.0 GENERAL DESIGN CRITERIA

5.1 Regulations

The electrical design will comply with all statutory requirements promulgated in the applicable Senegal Laws and Regulations. It will be designed to comply with all mandatory requirements and conditions established in the relevant Grid Codes and by the Electrical Grid Operator by SENELEC.

5.2 Standards

Electrical equipment, materials, components, and functions will be designed, built, rated, tested, and perform in accordance with the standards published by the following organizations.

IEC	International Electro technical Commission
IEEE	Institute of Electrical and Electronics Engineers
ISO	International Standards Organization
EN	European Standards
NEMA	National Electrical Manufacturers Association
NFPA	National Fire Protection Association

5.3 Redundancy

Equipment of the electrical system that is essential to the plant availability will have redundant design.

All the auxiliary transformers and service transformers will have 2X100% configuration except ACC service Transformer which is in 3 X 50% configuration.

All the MV Switchgear and LV Switchgear will have redundant supply feeders.

Station Battery (125V DC) will have 2X100% configuration for Main Plant.

Battery chargers (125V DC) will have 2X100% configuration for Main Plant.

UPS Inverters will have 2X100% configuration for Main Plant.

Station Battery (125V DC) will have 2X100% configuration for 225kV GIS.

Battery chargers (125V DC) will have 2X100% configuration for 225kV GIS.

UPS Inverters will have 1X100% configuration for 225kV GIS.

Generator step-up transformers will have 3 X 50% radiator banks.

5.4 Operating environment

As per WORLD METEOROLOGICAL ORGANIZATION's World weather information service, Climatological information at Dakar, (based on monthly averages for the 30-year period 1981-2010) is given below:

Month	Mean Daily Minimum Temperature (°C)	Mean Daily Maximum Temperature (°C)
Jan	17.9	24.9
Feb	17.8	24.8
Mar	18.3	25.1
Apr	19.1	24.7
May	20.6	25.9
Jun	23.5	28.5
Jul	25	29.9
Aug	25.2	30.2
Sep	25	30.6
Oct	25.2	30.9
Nov	23.2	29.7
Dec	20.5	27.2

It is noted that Mean daily maximum temperature is not more than 30.9°C.

For the electrical equipment connected in the GTG /STG train, where output varies with ambient air temperature, appropriate temperature will be considered for the applicable output rating.

For other electrical equipment, design Ambient Air Temperature of 40°C will be considered for indoor & outdoor installations. Design Ambient Air Temperature of 30°C will be considered for Electrical equipment located in air-conditioned rooms. Design ambient ground Temperature of 30°C will be considered for below ground cabling installations.

Equipment installed in the vicinity of gas and liquid fuel lines will be classified according to IEC hazardous classification and equipment installed within the classified zones will be certified accordingly.

Electrical equipment and cables will be de-rated according to Vendors' recommendations for the applicable laying conditions and ambient temperature.

5.5 Ingress Protection

Electrical equipment enclosure will have following ingress protection:

- Gas Insulated Switchgear : IP65
- Isolated phase bus duct : IP55
- Indoor MV Switchgear : IP4X
- NGR Cubicle (Outdoor) : IP55
- Indoor LV Switchgear : IP42
- LV Distribution Board : IP31
- Outdoor LV Switchgear : IP55
- Motors : IP55



5.6 Labelling

KKS numbering will be implemented for equipment and systems.

Electrical panels will be provided with danger signage advising of voltages employed within.

All safety signage will be in both French and English.

5.7 Paint shade

The following paint shades are proposed for the Electrical equipment

S.No.	Equipment	Colour	RAL no.	Shade
1.	Motor	Capri Blue	5019	
2.	Busduct	Pastel Turquoise	6034	
3.	Transformers	Pebble grey	7032	
4.	MV & LV Switchgear	Manufacturer Standard /Pebble grey	7032	
5.	Control & relay panels	Manufacturer Standard /Light grey	7035	



6.	Battery charger, UPS cabinets	Manufacturer Standard /Light grey	7035	
7.	GIS	Agate Grey	7038	
8.	NGR	Pebble grey	7032	
9.	BSDG	Manufacturer Standard		



6.0 POWER SUPPLY SYSTEM DESIGN CRITERIA

6.1 Power Generation

Gas Turbine generators are supplied by GE. The generator is rated for 129.2 MW (@ 40 ° C cold gas temperatures) at 0.85 pf, 15 kV, 3000 rpm, 50 Hz. It is totally enclosed with water to air heat exchanger (TEWAC) cooling system. It will have brushless excitation system. Generator will be capable of delivering rated output from gas turbine in the entire operating range/temperature.

At lowest ambient of 10 °C, GT output is 134.14 MW. Generator is rated to carry > 144 MW at lowest ambient of about 15 °C.

Steam Turbine generators are supplied by GE. The generator is rated for 144.5 MW (@ 40 ° C cold gas temperatures) at 0.85 pf, 15 kV, 3000 rpm, 50 Hz. It is totally enclosed with water to air heat exchanger (TEWAC) cooling system. It will have brushless excitation system. Generator will be capable of delivering rated output from steam turbine in the entire operating range/temperature.

At lowest ambient of 10 °C, ST output is 143.14 MW. Generator is rated to carry > 154 MW at lowest ambient of about 15 °C.

Normal operation of the plant (Normal start, load control and shut down) to be performed directly from DCS/PCS. The HMI pages of subsystems (GTG, STG, GIS etc.) shall be fully mirrored in DCS to prevent operator's confusion.



6.2 Power Evacuation

Power evacuation is envisaged at two levels, one at Future 90 kV through existing GIS substation and the other at 225kV level through 1 no. of outgoing Overhead line.



The Existing 90 kV GIS (Model B65, GE make) is at a distance of 300 M from the proposed power plant. It has double bus configuration and will be extended with Two (2) Generator Transformer bays.

Salient parameters existing 90 kV GIS are given below:

- Maximum Voltages 145 kV, 50 Hz
- Short-duration power-frequency withstand voltage 275 kV
- Lightning impulse, phase-to-earth / across isolating distance 650 kVp
- Continuous current rating for Feeders 1250 A
- Continuous current rating for Bus 2500 A
- Short-time withstand current 40 kA



- Peak withstand current 100 kAp
- Duration of short-circuit 1s
- Installation indoor

The new 225kV GIS substation will have double bus configuration with the following bays:

- Three (3) Generator Transformer bays
- One (1) spare Transformer bay
- One (1) Line bays
- Three (3) Spare Line bays
- One (1) Bus Coupler bay
- Two (2) Bus VT bays



Salient parameters 225kV GIS are given below:

- Maximum Voltages 245 kV, 50 Hz
- Short-duration power-frequency withstand voltage 460 kV
- Lightning impulse, phase-to-earth / across isolating distance 1050 kVp
- Continuous current 2500A
- Short-time withstand current 40 kA
- Peak withstand current 100 kAp
- Duration of short-circuit 1s
- Installation Indoor

6.3 Auxiliary power supply

Auxiliary power distribution system is envisaged to feed the plant auxiliaries. To feed motors rated ≥ 200 kW, 6.6 kV supply is envisaged and for lower rating motors, 400 V supply is envisaged. In the power plant, HRSG feed pump, Gas compressor motor and GT cranking motors are rated ≥ 200 kW, for which 6.6 kV supply is required. Hence Auxiliary power supply is envisaged at 6.6 kV and 400 V level.

Generator Circuit Breaker (GCB) scheme is envisaged for the plant. For feeding the Unit and Station auxiliary loads, 2x100% rated Unit auxiliary transformers are envisaged for GTGs. Each Unit auxiliary transformer is tapped off from the Main run off IPBD between GCB and Generator Step-up Transformer

6.6kV supply will be sourced from 15/6.9 kV Unit Auxiliary transformers. The low voltage supply will be sourced from 6.6/0.42 kV Service transformers.

The following voltage levels are envisaged:

- | | |
|---|---|
| MV voltage | 6.6 kV +/- 5 %, 3-phase, resistance earthed neutral |
| Motors 200kW & above | 6.6 kV +/- 10 %, 3-phase, (L1, L2, L3) |
| Motors from 0.25 to and < 200kW | 400 V +/- 10 %, 3-phase, solidly earthed neutral (L1, L2, L3, PE) |
| Motors less than 0.25kW | 230 V, 1-phase, solidly earthed neutral |
| Lighting and other loads | 400/230 V, 4-wire, 3-phase + neutral, solidly earthed neutral |
| AC control supply & Space Heater supply | 230 V, 1-phase, solidly earthed neutral |

Key Single line diagram indicating Power evacuation and standby power supply arrangement is indicated in the drawing (DWG.No. SNC-36-AB-EDU-CEI-001).

6.4 Black start DG system

Black Start Diesel Generators (BSDG) is envisaged to provide start up power supply for the Gas turbine unit. BSDG will also provide power supply for safe shutdown of the turbine generators in the event of total grid failure and GTGs are also tripped and will also provide emergency backup power for critical items such as the Battery chargers, UPS, lube oil pumps, air compressors control room lights, and DCS supply.

BSDG will be suitable for starting the GT cranking motor. The Cranking Motor will be started through Auto transformer. Four numbers of 1250 kVA DG sets are envisaged.

6.5 125 V DC supply

The control voltage for Gas turbine system will be 125 V dc supplied from station batteries.

The control voltage for MV and 400V Switchgear circuit breakers protection circuits, tripping, closing, and spring charge motors will be 125 V dc supplied from station batteries

The control voltage for 225kV GIS System, Control & Relay equipment will be 125 V dc supplied from 225 kV GIS Batteries.

The control voltage for 90kV GIS System, Control & Relay equipment will be 127 V dc supplied from existing 90 kV GIS Batteries.



6.6 UPS Supply

Uninterruptible power supplies (UPS) will be used to provide AC power to all critical central control room equipment and nominated key equipment of the plant process control systems, i.e. SCADA systems, network communications equipment, PLCs, etc.

6.7 Neutral grounding

6.6kV voltage level is grounded through the neutral grounding resistors, limiting the fault current to about 300A.

400V system will be solidly earthed.

225 kV & 90 kV system will be solidly grounded.

15 kV system will be grounded through Neutral grounding transformer.

7.0 EQUIPMENT DESIGN CRITERIA

7.1 Generator Step-up Transformer

Generator Step-up transformer (GSUT) will be provided for each generator to step up the generating voltage for connection to the 225kV & 90kV grid. The Transformer will be sized to carry the maximum generator output of its relevant turbine output at all operating conditions / ambient conditions of its relevant turbine. The winding material shall be copper.

For Gas turbine, at lowest ambient of 10 °C, GT output is 134.14 MW and MVA rating is 157.8 at 0.85 PF. Generator step-up transformer in the range of 180 MVA will be required (@ yearly weighted average temperature of 20 °C) .

At lowest ambient of 10 °C, ST output is 143.14 MW and MVA rating is 168.4 at 0.85 PF. Generator step-up transformer in the range of 180 MVA will be required (@ yearly weighted average temperature of 20 °C) .

Generator Step-up transformer will be 3 phase construction, bi directional power flow; mineral oil filled with oil natural air forced (ONAF) cooling to carry its 100% rated power. ONAN rating will be 80% of ONAF rating.

GSUT will be provided with 3x50% radiator banks.

Salient parameters of GT – GSUT is as follows:

- HV1 winding : 180 MVA, 225 kV, with OLTC $\pm 10\%$ in steps of 2.5%, Solidly grounded,
- HV2 winding : 180 MVA, 90 kV with OLTC $\pm 10\%$ in steps of 2.5%, Solidly grounded
- LV winding : 180 MVA, 15 kV
- Vector group : YNYNd11

Salient parameters of ST – GSUT is as follows:

- HV winding : 180 MVA, 225 kV, with OLTC $\pm 10\%$ in steps of 2.5%, Solidly grounded,
- LV winding : 180 MVA, 15 kV
- Vector group : YNd11

7.2 Unit auxiliary transformer (UAT)

The unit auxiliary transformer is teed-off from the main connections of the Gas turbine generator to the generator transformer. 2 X 100% Unit auxiliary transformers will be provided to feed plant auxiliaries. Each UAT shall be rated to feed entire plant auxiliaries.

Unit auxiliary transformer will be 3 phase 2 winding type. HV side voltage will be generator voltage and LV side will be connected to 6.6kV system. No load LV voltage will be 6.9kV to take into account voltage drop during loaded condition.

Unit auxiliary transformer will be mineral oil filled with oil natural air forced (ONAF) cooling to carry its 100% rated power. ONAN rating will be 80% of ONAF rating.

Unit auxiliary transformer will be provided with tank mounted radiator banks of 2x 50% capacity.

HV winding will be delta connected and LV winding will be star connected. LV neutral will be grounded through neutral grounding resistor to limit the earth fault current to about 300A.

The transformers will be provided with off circuit tap changer provided on HV side with operating range of $\pm 2.5\%$ in steps of 1.25%.

7.3 Generator circuit breaker (GCB)

Generator circuit breaker will be provided in each Generator circuit, connecting between generator and associated Generator Transformer. GCB will be kept opened during start up and will be closed when generator builds up rated voltage and synchronized with the grid using the GCB.

GCB will consists of SF6/Vaccum Circuit Breaker chamber, Line disconnecting switch, Earth switch, Surge arresters, voltage transformers and Manual short circuiting bar.

GCB will be rated for 15 kV, sized to carry the full load current of Generator at its entire operating range. 10% design margin will be considered in arriving at the current rating.

The short time current rating of the GTG-GCB shall be decided considering higher of the short circuit current contribution from generator or the short circuit current contribution from 225/90 kV system through generator transformer and contribution of MV auxiliary system through unit auxiliary transformers. GTG-GCB is supplied by GE.

The short time current rating of the STG-GCB shall be decided considering higher of the short circuit current contribution from generator or the short circuit current contribution from 225 kV system through generator transformer. STG-GCB is supplied by GE.

7.4 Isolated phase bus duct (IPBD)

For connection between generator line side MV Cell / GCB, Generator stepup transformer and unit auxiliary transformers; isolated phase bus duct will be provided. Bus duct will be of continuous type. The bus duct will be with Copper conductor and aluminum enclosure.

IPBD main run will be rated for 15 kV, sized to carry the full load current of Generator at its entire operating range. 10% design margin will be considered in arriving at the current rating.

IPBD tap-off to UAT will be rated for 15 kV, sized to carry the full load current of UAT. 10% design margin will be considered in arriving at the current rating.

The short time current rating of the GTG main bus duct will be arrived at based on the higher of the following two values:

- Short circuit current contribution from generator plus contribution of 6.6kV auxiliary system through unit auxiliary transformers; (OR) short circuit current contribution from 225kV/90kV system through generator transformer plus contribution of 6.6kV auxiliary system through unit auxiliary transformers.

The short time current rating of the STG main bus duct will be arrived at based on the higher of the following two values:

- Short circuit current contribution from generator (OR) short circuit current contribution from 225kV system through generator transformer.

The bus duct will be natural air-cooled and the busbar conductors will be capable of withstanding temperature of 90°C for plain bolted joint, 105°C for silver plated bolted joints and aluminum welded joints.

7.5 6.6 kV switchgear

6.6kV Switchgear will be of Indoor, single front, single tier, metal clad, fully draw out type with SF6 type circuit Breaker.

The switchgear will be sized to carry the full load current of UAT. 10% design margin will be considered in arriving at the current rating. Based on the fault level, the short circuit withstand rating shall be rated for 31.5 kA for 1 sec. One minute Power Frequency withstand voltage will be: 20 kV (rms) and basic impulse withstand voltage will be 60 kVp.



7.6 LV service transformers

LV Service transformers for the Plant (Except ACC) will be provided to feed the loads at 400 V levels of auxiliary power systems, with the criteria that each switchgear will be fed by two numbers 100 % rated transformers. For ACC Service, During normal operation Transformer#1 will feed ACC MCC Switchgear#1 and Transformer#2 will feed ACC MCC Switchgear#2. Transformer#3 will remain charged and in standby mode. In case of failure of supply on any one MCC (U/V) and without any protection trip operated; Standby incomer fed by Transformer#3 of that particular MCC will be closed will start feeding the loads.

Each transformer will be rated to meet the loads connected on both the bus section of Switchboard with 10 % design margin.

Maximum rating of the transformer will be limited to 2.5 MVA, so that short circuit rating of 400 V system is not exceeding 50 kA except for ACC Transformers which will have short circuit rating not exceeding 65 kA on 400 V system. Further, Transformer voltage impedances will be so optimized that the auxiliary system voltages under various loading conditions and fault currents are always within permissible limits and equipment are not subjected to unacceptable voltages during operation and starting of motors. 

HV winding will be delta connected rated for 6.6 kV and LV winding will be Star connected rated for 400 V, except for BSDG where HV winding will be Star connected for 6.6 kV and LV winding will be Delta connected rated for 400 V. LV neutral will be solidly grounded. No load voltage ratio will be at least 5% higher than the nominal voltage to account for voltage drop in the transformer during loaded condition. Accordingly LV voltage will be 420 V.

Service transformer will be 3 phase 2 winding type Hermetically sealed oil filled ONAN outdoor type. The transformers will be provided with off circuit tap changer provided on HV side with operating range of $\pm 5\%$ in steps of 2.5%. 

7.7 LV Busducts

Non Segregated Phase Busduct is provided for interconnection between ACC Service Transformer-1,2&3 and ACC Switchgears; Unit Service Transformer-1&2 and Unit Service Switchgear; Station Service Transformer-1&2 and Station Service Switchgear and BSDG Transformer-1&2 and BSDG Switchgears. For others, cables will be provided. 

Current rating will be selected to carry transformer rating with 10% design margin. Size of the neutral bus conductor (Except for BSDG) shall be not less than 50% of size of the phase conductor. Bus duct will be rated for short circuit withstand rating 50 kA for 1 sec except for Bus duct interconnected between ACC Switchgear and transformer which shall be 65 kA for 1 sec.

The bus ducts will be non-segregated type with Copper conductor. The enclosure will be of Cast resin. The Bus duct will be natural air-cooled suitable for outdoor duty.

7.8 LV Switchboards

Power Control Centre (PCC)/Power Motor Control centre (PMCC) will have two bus sections and a bus coupler. Each section will have 100% rated incomer fed from service transformer. The PCC/PMCCs will distribute power to the following

- To motors rated up to 200KW
- To motor control centers (MCC).
- To Distribution Boards
- To Outgoing feeders



The MCCs will distribute power to the following

- To LT motors <75 kW.
- To outgoing feeders.

The Motors of rating less than 0.375kW shall be feed from panel boards.

The 400 volt Distribution boards will distribute power to small 400/230 V non motor loads.

The continuous current rating of the PCC/PMCC buses and their incomer breaker will be the rated for secondary winding current of their upstream transformer with 10% margin.

The normal operating conditions for the PCC/PMCC & MCC will be with both the incomers closed and bus coupler open. The incomers are sized such that one incomer can supply power to both the buses through the bus coupler, when one of the incomer is out of service.

Redundant mechanical loads will be fed from separate PMCC/MCC buses.

PCC/PMCC/MCC/DB will be rated for 400 V, 50Hz, 3-phase, 4 wire supply except for BSDG switchgear which is 3-Phase , 3 wire supply.



PCC/PMCC & MCC will be form 3B, fixed type and distribution boards will be fixed type. All MCC & Floor Mounted DB's shall be of double front type.

All the motor starters will be 'Direct On Line' type except for ACC motors which will be star delta type. All motor starters will confirm to Type 2 co-ordination. Motor feeders will be as detailed below -

- Motors up to 200 KW are controlled by "MPCB or MCCB- Contactor - Electronic Overload Relay with current display unit" - " combination

The continuous current rating of the bus bars, incomers, bus couplers of the MCC will be the maximum load on the bus due to all the running auxiliaries during any operating condition plus 10% margin and the biggest spare feeder rating rounded off to the next higher standard rating.

The PCC/PMCC, MCC and Distribution boards will be located indoors. Type of outgoing feeders will be as follows:



- Motors rated up to 200 KW : MPCB or MCCB Contactor Controlled.
- Feeders rated > 800A & Above : Air circuit breaker (ACB) controlled.
- Feeders rated $\leq$ 800A : MCCB controlled.

Each PCC/PMCC will be provided with auto changeover scheme and manual live changeover scheme for planned changeover between incomers and bus coupler.

7.9 Battery & Charger

125V DC Power Supply System will provide normal source of power to the 125V DC loads, such as the emergency oil pumps, Control Supply to Switchgears / Panels and emergency lighting. It will

also be capable of constantly supplying emergency loads for a period of 2 hours when AC supply to Charger fails.

For each Gas turbine unit, one set of batteries and 2X100% float cum boost charger will be provided. 125 V DC system for gas turbine units will be provided by GE. 125 V DC of both the GTs will be interconnected for backup supply.

For STG DC loads and station DC loads, a dedicated 125 V DC system will be provided. 2 x 100% battery and 2 x 100% float cum boost charger will be used. Each battery and battery charger will be sized to cater to 100% loads of STG and station loads.

For 225kV Switchyard DC control, a dedicated 125 V DC system will be provided. 2 x 100% battery and 2 x 100% float cum boost charger will be provided.

For 90kV Switchyard control, 127V DC supply from existing DC system will be provided.

The 125V DC power supply system will be ungrounded system.

Lead acid VRLA type battery is proposed. The battery will be sized with design margin, temperature correction factor and an ageing factor in accordance with IEEE standard.

7.10 UPS Inverter

To provide Uninterrupted power supply to DCS/PCS, STG loads, station loads and other Control Panels, 400 V AC UPS system is envisaged.

For each gas turbine unit, UPS supply will be derived from Gas turbine DC system through inverter. Standby supply will be provided from Station UPS.

UPS loads for Steam turbine and station loads like DCS, etc. will also be fed from Station UPS. The Station UPS will be fed from 125 V Station DC system through 2x100% Inverters with 50% load sharing facility. Station UPS will have servo controlled voltage stabiliser to provide backup supply.

7.11 Black start DG SET

Diesel generator sets (DGs) are proposed for Black start of any one Gas Turbine unit, in the event of failure of Grid supply and also for feeding emergency loads for safe shut down of the Unit.

DG sets of 1250kVA will be rated for Prime duty. Four numbers are envisaged for Plant.

DG sets will be sized to carry 100 percent of the emergency/ Black start loads with a design margin of minimum 10% considered in DG sizing.

Alternator will be rated of 400 V, 3 phase, 3 wire supply.

Each DG set will be provided with acoustic enclosure for outdoor installation

DG sets will have AMF (Automatic Mains Failure) start facility and will be capable of starting and picking up the loads. Start / stop, voltage raise / lower, speed raise / lower controls are envisaged from AMF panel.

All the four DG sets will be design for continuous parallel operation.

7.12 Motors and actuators

All AC Motors will be squirrel cage induction type rated for continuous duty. DC Motors will be shunt wound type. Power supply for AC motors will be as follows :

- Below 0.25KW : 230 V, 1 Phase, 50 Hz
- 0.25KW to < 200 kW : 400 V, 3 Phase, 50 Hz
- $\geq$ 200 kW : 6600 V, 3 Phase, 50 Hz

Motors will be capable of delivering the rated output with supply voltage variation of $\pm 10\%$ and frequency variation of $+3\%$ to -5% and absolute sum of 10% . DC Motors will be rated for 125 V DC with supply voltage variation of -15% to $+10\%$.

LT Motors will be considered of energy efficient type (IE2). All the motors will be suitable for 'Direct On line' starting except for ACC Fan motors which will be star delta type starting unless otherwise specified.

The Motors greater than 75kW and less than 200kW are connected to station service LV Switchgear. However the Motors 75kW and below are connected to station service LV motor control center. Where as 0.375 kw and below shall be feed from panel boards.



While sizing the motor, deration due to voltage and frequency fluctuation need to be accounted. A design margin of minimum 10% will be considered over the maximum input power requirement of the driven equipment at rated point in Motor sizing unless otherwise specified in the Mechanical system criteria.

Motors will be capable of starting and accelerating the load with direct on line starting without exceeding acceptable winding temperatures when supply voltage is 80% of the rated voltage.

Motors will be suitable for the following starts under the specified conditions of load, torque and inertia in the motor initially at its normal operating temperature.

- No. of consecutive cold starts : 3
- No. of consecutive hot start : 2

All motors will have IP 55 degree of protection. Motor for outdoor or semi-outdoor service will be of weather-proof construction. Motors located in hazardous area will have flame proof enclosure as per applicable standard.

Type of enclosure/cooling for LT motors will be TEFC and for HT motors CACA/TETV/TEFC.



All insulated winding will be of copper. All the motors will have class F insulation. Temperature rise will be limited to Class B.

Motors rated 1000 kW and above will be provided with differential protection. Required current transformers, one for each phase will be mounted in the neutral side terminal box.

"Anti-condensation heaters shall be provided for all motors rated above 50 kW. The heaters shall be automatically controlled from the motor starter and energised when the motor is shut down. "

Actuators will be of integral type. Starter module required for the actuators will be provided in actuator itself (Integral to actuators). Actuators will be with IP 65 enclosure. Actuator motors will be rated for short time duty / intermittent duty to suit the valve application. Actuator Motors will be suitable for direct on line starting.

Motorized actuators will be provided with removable plug and socket arrangement for both power and control cable for ease of isolation during valve maintenance and actuator removal.

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7.13 Variable Frequency Drives

Motors required to operate at adjustable speed will be controlled by electronic AC adjustable frequency drives. Adjustable frequency controllers larger than 30 kW will be complete with input filters. Output filters may be considered on the drives where the length of the cable is considerable and requires further attenuation.

Drives up to 30kW will generally be installed in the low voltage MCC. Larger drives will be either installed in separate drive suites or be free standing.

The use of Adjustable frequency drives is to be avoided, if possible.

7.14 EHV Cables

225 kV EHV Power cables will be provided for inter connection between Generator Step-up transformer and GIS Switchyard (New 225kV and Existing 90kV Switchyards). Cables will be sized to satisfy the following Criteria:

- Short circuit withstand capacity for applicable fault current and duration.
- Full load current carrying capacity under installation conditions considering Site ambient temperature & site installation (Grouping) conditions based on Manufacturer's recommendation.

Short circuit withstand duration for conductors will be not less than 1 sec. Cable screen will be sized based on calculated system fault level considering non-adiabatic effect as per IEC 60949.

Cables will be 225 kV rated, Unarmoured type with copper conductor and XLPE insulation.

EHV Cables outersheath shall of Black colour.

The cables will conform to the relevant IEC standard and will constitute the following:

- Circular Stranded Compacted Annealed Copper (Class 2) [Water Blocked] Conductor
- Semi-conductive Water Blocking Tape
- Extruded Thermosetting Semi-conducting Compound as Conductor Screen
- XLPE Insulation
- Extruded Thermosetting Semi-conducting Compound as Insulation Screen
- Semi-conductive Water Blocking Tape
- Metallic Screen of Welded Corrugated AL Tape
- HDPE (ST7) Outer sheath
- Graphite Coating



7.15 Power & control cables

Power cables will be sized to satisfy the following Criteria:

- Short circuit withstand capacity for applicable fault current and duration.
- Full load current carrying capacity under installation conditions considering Site ambient temperature & site installation (Grouping) conditions based on Manufacturer's recommendation.
- Permissible voltage drop limits under steady state/transient state as applicable.

For breaker operated feeders, Short circuit withstand duration for conductors will be not less than 160 msec for motor / transformer feeders, 500 msec for tie feeders and 1 sec for incomers in general. For 6.6 kV cables, screen will be sized to withstand 300 A for 3 sec.

Voltage drop from transformer secondary to motor terminals during full load running of motors will be limited to 3% of rated voltage. Voltage drop from transformer secondary to motor terminals during starting of motors will be limited to 8% of the rated voltage for all motors except ACC Fan Motors which will be limited to 10%.

All cables will be provided with copper conductor. The Minimum power cable size shall be 2.5sqmm. For CT/CVT/PT the minimum cable of 2.5sqmm; for Controls & Panel internal wiring minimum 1.5 Sqmm shall be considered.

EHV Cables & 6.6 kV power cables will be XLPE insulated. LV Power cables will be XLPE insulated. Control cables will be XLPE insulated for GTG Area for PVC insulated for BOP Area.

PVC insulation will be suitable for continuous conductor temperature of 70°C and short circuit conductor temperature of 160°C. XLPE insulation will be suitable for continuous conductor temperature of 90 °C and short circuit conductor temperature of 250 °C.

Cables for EHV system will be rated for 76/132(145)kV for 90kV and 130/230(245)kV grade for 225kV. Cables for 6.6 kV system will be rated for 6/10kV grade. Power Cables for 400/230 V AC and 125V DC systems will be rated for 0.6/1.0 kV grade.

The cables will conform to the relevant IEC standards with FR properties for the outer sheath. MV Cables outersheath shall of Red colour. LV Power Cables outersheath shall be of Black Colour. Control Cables shall be of Grey Colour.

7.16 Cabling System

EHV cables, MV Cables, LT Power cables, Control cables >60V and I&C cables (Control Cables & Signal Cables <60V) will be laid in separate trays and different type of cables will not be mixed on the same tray.

In case of horizontal formation, the highest voltage cables will be laid in the bottom most position in the tray stack followed by other voltage grades in the descending order (i.e MV Cables in Bottom Tray & Signal Cables in Top Tray).

Power & Control cables will be laid on ladder type trays. Control and Instrumentation & Signal cables will be laid on perforated type trays. Cable trays will be supported at an interval of 1500 mm. Vertical runs will be supported at an interval of 1000 mm. Single core cables will be laid in trefoil formation using trefoil clamps at a spacing of 2000 mm, however spacing shall be decided as per the clamp strength as per IEC 61914.

Cable trays will be of prefabricated galvanized steel. Cable tray support system will be of prefabricated, galvanized steel bolted type.

Cable laying criteria for the cables as follows

- For MV cables, Single core cables will be used in trefoil formation between each trefoil formation one cable diameter spacing will be kept maximum fill rate will be 50%.
- For LV cables, flat formation will be followed with single layer. if any space restriction, double layer will be used. Maximum fill rate will be 60%.

- For LV small power installation/lighting/sockets/mov/etc cables, several layers with the filling ratio 60%.
- For LV Control cables, several layers with the filling ratio of 70%.

PVC conduits embedded in concrete duct banks shall be used to route the cables. Cable duct banks, the conduits embedded into concrete or buried into block works shall be of PVC type as per the standard "TS EN ISO 1452-2".

For Conduits which are buried directly in ground, HDPE pipes shall be provided as per the standard "TS EN ISO 1452-2".

7.17 Illumination system

The plant lighting system includes

- Normal AC lighting and emergency AC lighting which contributes together 100% lighting
- Emergency DC lighting in selected areas of the plant during plant emergency conditions.

Normal AC lighting will comprise of lighting transformer, main lighting distribution board (MLDB) and lighting panels (LPs). Each LP will consist of outgoing circuits supplying power to lighting fixtures. LPs will be supplied by MLDB. MLDB supply (400V, 3 phase, 4 wire) will be derived from delta / star lighting transformer with effectively earthed neutral. For main plant building, lighting transformer and MLDB will be fed from 400V Unit PCC/PMCC. For auxiliary plants far away from the main plant, lighting distribution boards will be fed from respective 400V PMCC / MCC.

LED type of fixtures will be used for reduction in energy consumption, long life and robust construction. Minimum types of LED luminaires have been utilized in order to reduce the varieties of inventory and spares for maintenance.

LED Luminaires will be used for the lighting of all the indoor & outdoor areas. In false ceiling area LED luminaires will be recessed mounting type & in non-false ceiling area the LED luminaires

All LED luminaires will be rated L70 with a minimum of 50,000 hours and colour rating of 4000K.

To enable switching off lights when they are not required normal lighting fixtures will be provided with local isolating switches. Exit light fixtures will not have local isolation.

General lighting for open areas within non-hazardous areas will be by means of LED floodlight luminaires mounted on adjacent structures or on strategically located floodlight columns. High mast floodlighting installations will also be used where appropriate.

Road lighting poles (8M) / Flood lighting poles (High Mast) (18M) will be of painted/polygonal galvanised steel pole. The street light/Outdoor fixtures shall be of control by manual and Photo sensor including timer.

Lighting level design will include a Maintenance factor to account for lamp lumen depreciation, luminaires surface dirt and room surface dirt, etc.

7.18 Lux level will be maintained as per 11.BOP Electrical_Rev02" document. Lux level for various areas are given below:

- | | |
|---|-----|
| • Turbine Hall | 200 |
| • HRSG Platforms | 100 |
| • Switch gear /MCC rooms, Battery room, DG Room | 200 |

• Main Control room	500
• Control equipment room	300
• Office area	300
• Air compressor house	150
• Pump houses, WTP area	150
• Transformer yard	30
• Roads	20
• Cable galleries	100
• Stair cases, Passages, Toilets	100
• External working area with presence of machinery	100
• Warehouse	150
• Workshop	300
• Laboratory	150
• Cafeteria	100
• GIS area	100

7.19 Earthing & lightning protection system

The earthing system will be designed to ensure that the earth resistance of the interconnected mats will be less than 1 ohm. 

Earth mat design will be carried out as per standard IEEE: 80. The earthing system will meet the following requirements:

- Provide low impedance earth fault current return path for operation of earth fault protection / alarm.
- Withstand maximum earth fault current for the fault duration
- Limit step and touch potentials within acceptable levels for areas where earth fault current return path is through earth (eg. switchyard)

Earthing design will depend on soil resistivity value. The Average soil resistivity is 55 ohm meter for Power house area and 25 ohm meter for GIS substation. Soil resistivity also determine rate of corrosion for buried earth conductors. The following norms, are used in determining corrosion allowance for buried earth conductor. 

- Soil resistivity greater than 100 ohm-m : No allowance

Based on final soil resistivity values, corrosion allowance will be considered in determining the buried conductor cross section.

Earth mat will be sized for actual fault current (as per short circuit analysis report) for duration of 0.5 second. Buried earthing conductor will be of bare copper wire. Equipment earthing conductor installed above ground will be of Copper flat/insulated copper conductor.

The Earth mat will be buried in the ground at a minimum depth of about 600mm in a mesh form.

For neutral earthing of each generator & transformer, 2nos. treated earth pits will be provided. 

Electronic earthing (DCS and PLC systems) will be earthed utilizing an insulated ground wire to a separate earth system.

Protection against lightning strokes will be provided for power plant. The Lightning protection system will have horizontal / vertical air terminations on roof top, down conductors, test links and

earth connections as per IEC 62305. Each down conductor will be connected to lightning test link which also is connected to a grounding rod which will be further connected to main earth grid.

7.20 **Plant communication System:**

The following communication system will be provided:

- Public address system with page and party line
- Telephone system with 10 external lines and 50 intercoms

8.0 **CONTROL & PROTECTION OF ELECTRICAL SYSTEM**

8.1 **Control Aspects**

Each Gas turbine unit's electrical system will be controlled from GE controller located in each PEEC cubicle. GE controller will be interconnected with Plant DCS.

Steam turbine generator will be controlled from Unit Control panel provided by STG Supplier.

Main Plant Electrical System will be controlled from DCS.

In DCS, the controls will be through mimic screens in the VDU. Separate mimic screens will be provided for each Switchboards/System.

Digital signals, Analog signals, Annunciation and controls of the switchboards will be extended to DCS system. Sequence event recording / status of breakers & fault reporting of Electrical system.

For all motors, Local Pushbutton station (LPB) with latching Emergency Stop pushbuttons will be provided. Start push button will be provided based on process/control requirement.

GIS Substation will be controlled from SCADA. But, links will be provided to DCS system to enable monitoring of GIS substation.

Generally direct-on-line (DOL) motor starting will be used for all motors except those that require variable speed drives. Star-Delta starters may be considered for drives above 55 kW where excessive voltage drop is a consideration. Voltage drop during motor starting should be kept to a maximum of 8% except ACC fan motors which are with 10%.

Starting, stopping and coordination of motor drives will be from the DCS control system via either a control and status communication link to the electronic motor protection relay, or by hard wire connections.

Emergency stop pushbuttons will be provided near the motors.

8.2 **Synchronizing Facilities**

Automatic synchronisation units of the numerical type will be provided to synchronise the generators via the generator breaker or the HV circuit breaker in the grid substation as the case may be.

Synchronizing panel will be located in Main Control Room and will be part of GE supplied Control Panel. For GIS, the same will be part of GIS Vendor scope and will be located in Switchyard Control room.



The synchronisation process will be initiated by the start-up program of the DCS or by manually given operator command after the desired synchronisation point (i.e. HV CB or Generator CB) has been selected.

The synchronisation unit will automatically check whether the necessary conditions are being met, i.e. matching phase angle, voltage and frequency on both sides of the open breaker.

The voltages, frequencies, phase angles, and the status – ON-FAULT-BLOCKED – of the synchronising unit will be indicated in the CCR.

8.3 Protection System

Fully graded protection system with requisite speed, sensitivity and selectivity will be provided for the entire Station.

For Generators GE will provide the protection system.

For generator transformer, unit auxiliary transformers (UATs) protections will be connected to two independent groups/relays, such that one protection system will always be available for any type of fault.

All the protection relays will be multifunction numerical type with self monitoring and diagnostic features and communication link/port.

Accuracy class of CT core meant for tariff Metering/Energy auditing at 225kV/90kV level will be 0.2S. Accuracy class of CT core meant for 6.6kV Level will be 0.5. Accuracy class of CT core meant for metering at 400 V level will be 1.0. Instrument security factor will be < 5. Accuracy class for Protection core will be 5P20 for protection in general. For differential protection, accuracy class for Protection core will be PX type.



Accuracy class of Voltage Transformers (VT) winding meant for tariff Metering/Energy auditing at 225kV/90kV level will be 0.2. Accuracy class of VT core meant for 6.6kV Level will be 0.5. Accuracy class of VT core meant for metering at 400V level will be 1.0. Accuracy class for Protection Core will be 3P.

8.4 Generator Step-up Transformer Protection

The GSUT Protection panel will have the following protections.



- Overall differential protection (87OA)
- GSUT differential protection (87T)
- HV REF protection (64R)
- Over fluxing protection (24)
- GSUT over current and earth fault protection (51N)
- Backup earth fault protection (51G)
- Transformer protection such as Buchholz, oil / winding temperature, pressure relief valve etc (63 / 49).

8.5 Unit Auxiliary Transformer Protection

The UAT protection panel will be provided with the following protections.

- Phase over current protection (50/51)
- Differential protection (87T)

- REF protection on LV side (64R)
- Backup earth fault protection (501G)
- Transformer protection such as buchholz, oil / winding temperature, pressure relief valve etc. (63/49)



8.6 MV Feeder Protection

The following protection will be provided for various feeders in the MV Switchgear.

Incomer feeder

- Synchro-check/voltage check (25)
- Three-phase under voltage 3 (27)
- Three-phase non-directional overcurrent 1 (50/51/51B)
- Fuse failure supervision (60)
- Three-phase directional overcurrent 3 (or non directional mode - 50/51) (67)
- Directional earth-fault 3 (or non directional mode - 50N, 51N (OR 50G, 51G)) (67N)
- Electrically latched lockout relay 2 (86)
- Restricted Earth Fault Protection (64R) (For Black Start DG Incomers only)



Bus Coupler (TIE Between Unit Switchgears)

- Synchro-check/voltage check (25)
- Three-phase under voltage 3 (27)
- Three-phase non-directional overcurrent 1 (50/51/51B)
- Fuse failure supervision (60)
- Three-phase directional overcurrent 3 (or non-directional mode - 50/51) (67)
- Directional earth-fault 3 (or non-directional mode - 50N, 51N (OR 50G, 51G)) (67N)
- Electrically latched lockout relay 2 (86)



Transformer Feeders

- Three-phase non-directional overcurrent 1 (50/51/51B)
- Three-phase directional overcurrent 3 (or non-directional mode - 50/51) (67)
- Directional earth-fault 3 (or non-directional mode - 50N, 51N (OR 50G, 51G)) (67N)
- Electrically latched lockout relay 2 (86)



Mechanical protection such as

- Buchholz,
- Oil / winding temperature
- Pressure relief valve etc.

Motor Feeders

- Three-phase under voltage (27)
- Three-phase thermal overload protection for motors 1 (49M)
- Three-phase non-directional overcurrent 4 (50/51/51B)
- Non-directional earth-fault 3 (50N/51N (OR 50G/51G))
- Locked rotor 1 (51LR)
- Motor start 1 (51MS)
- Cumulative start-up time counter 1 (66)
- Electrically latched lockout relay 4 (86)
- Stabilized three-phase differential protection for motors/generators 1 (87M/87G)



BUS PT

- Three-phase undervoltage 3 (27)
- Electrically latched lockout relay 4 (86)



8.7 LV Feeder Protection

The following protection will be provided for PCC incomer & bus coupler feeder and breaker operated MCC incomer & bus coupler feeders:

- 51 -Over current protection
- 51N -Earth Fault protection
- 27-Under voltage protection

For breaker operated motor feeders, numerical motor protection relay having comprehensive motor protection features described under MV motor protection will be provided.

MCCB feeders when provided will be with overload and short circuit protection as minimum.

8.8 Black start DG set protection

The following protection will be provided for DG set as minimum.

- Voltage controlled over current protection.
- Reverse power protection
- Standby earth fault protection.
- Restricted earth fault protection.
- Negative phase sequence protection.
- Field failure protection.
- Over / under voltage protection.
- Over / under frequency protection.
- Over load alarm

9.0 LAYOUT ASPECTS

Gas turbine and generator is located outdoor. MCC, GTG control panel, battery charger, Mark VI panels are located in outdoor cubicle PEECC. Battery for GTG is separately located in outdoor cubicle. These equipment are engineered by GE.

Electrical building will be provided near the STG building to accommodate electrical equipment of the STG & station electrical equipment. Cable spreader room will be provided for the switchgear room, while false floor is planned for control room / control equipment room. Main C&I Control room will be provided in Admin Building.

Generator step-up transformers & unit auxiliary transformers, will be located in the transformer yard.

6.6 switchgear, 400V PCCs/PMCCs & MCC, MLDBs, DBs, will be located in switchgear room of Electrical building. 125V battery & UPS battery will be located in a separate room in the Electrical building. Battery charger, 400V UPS inverter will be located in air conditioned atmosphere in control room floor.

LV service transformers will be located outdoor adjoining Electrical control building.



Minimum clearance of 800mm will be provided on rear of panel for single front boards. For double front boards, clearance from wall/column will be minimum 1500mm. For MV switchboards installations with two rows of boards facing each other shall be with minimum 2000mm for maintenance and operation. However for LV switchboards the same shall be minimum 1200/1500 mm based on the panel. Clearance between adjacent panels in a row will be $< 200\text{mm}$ or $\geq 800\text{mm}$.

A minimum clearance of 800mm will be provided between transformer and wall for LV transformers. For HV transformers the clearance will be higher based on maintenance requirements.

Oil draining facility will be provided for transformer if oil in one transformer or the aggregate capacity of the oil in all the transformer exceeds 2000liters. Draining pit below the transformer will accommodate at least 1/3 rd of the oil of the transformer.

Oil soak pit will be located at more than 5meters from the transformer. One common soak pit will be proposed for generator transformer sized to accommodate 110% of the volume of oil of the largest transformer.

Where oil capacity of the transformer (individual or aggregate) is $\geq 2300\text{liters}$, fire walls will be provided unless equipment / building are located at specified distance stipulated in regulations. Fire wall will be 355mm thick brick or 200mm thick RCC extended at least 600mm above the highest point of the transformer.

The batteries will be located in a Air Conditioned room.

The 225 kV GIS switchyard will be located in front of transformer yard. GIS building will have cable vault below GIS. RCC Cable trench will be provided for connection between GIS and generator step-up transformer.

For 225 kV GIS, a separate control room to accommodate relay panels, SCADA system, DC system, etc. will be provided

Cables will be laid on overhead cable trays in general. Indoor cable trenches can be considered based on equipment layout. In water treatment plants and fuel oil pump house usage of cable trench will be avoided, instead cables will be laid overhead trays. Outdoor concrete trenches will be minimised. Cable routing will be designed to avoid directly buried cables.

In cable trenches, distance between bottom most tier and bottom of trench will be 150 mm and clearance from top most tray to bottom of the trench cover will be 300 mm. A clear passage of 600 mm width will be provided in Cable trenches.

Vertical spacing between two tiers will be not less than 300 mm for all cable tray installations.

Overhead cable trays will be laid in horizontal formation in general.

In cable spreader room a clear access passage of at least 800 mm wide will be provided along the cable ways, for a clear height of not less than 1.8 M.

In Electrical Switchgear rooms, Fire barriers/ Fire stops will be provided for all direct cable entries into electrical Switchgear from Cable Vault. Further, Fire barriers/stops will be provided for fire rated wall/floor penetrations. Fire barriers/ Fire stops will provide a fire endurance rating of at least 2 hours.

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10.0 CATHODIC PROTECTION

The Protection shall be designed based on the National Association of Corrosion Engineers (NACE) and International Electrotechnical Commission (IEC)

The Cathodic protection shall be designed to protect each of the structures indicated for a minimum period of 25 years, with the exception of CPG for internal protection that shall be designed for a minimum protection of 10 years.

The Equipment and materials shall be designed , selected, and manufactured for continuous operation under the applicable site conditions.

The impact of electrical grounding system shall be considered during the design of the CPS.

All ground rods with in 3 mtrs of or electrically continuous with, a cathodically protected structure shall be insulated. The impact of 132kV and 33kV over head lines, running parallel to the buried pipeline, shall be duly considered evaluating and eventually neutralizing the effect of the induced current. A report of the relevant study shall be submitted at the beginning of the detailed engineering.

Cathodic Protection shall be designed to the following:

- 1) To Protect the underground pipes
- 2) Pipe Lines
- 3) Storage tanks Internal surface(Bottom and Walls up to 1 meter high).

Cathodic Protection May not be considered for

- 1) Reinforcing Bars
- 2) Concrete Structure
- 3) Foundations.

Calculation shall be carried about on impress current method. (Cathodic protection and required Calculation shall be done based on the requirements)

11.0 HEAT TRACING

Installation shall be to IEEE 515 Standard for testing, installation and maintenance of Electrical Resistance Heat Tracing

The maximum pipe wall temperature of the traced pipe shall not exceed the maximum continuous power off exposure temperature of the cable.

Splicing connectors shall provide a water proof connection and also a low resistance bond between the braid of the connected cables. Hot end termination shall be water proof.

Temperature Parameters:

The local thermostat controlling a heater cable shall be located such that the temperature sensed is representative of the total length of traced pipes.

In the case of freeze protection of water lines, several heater controllers controlled by single ambient- sensing thermostat set at 7°C shall be considered.

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12.0 AREA CLASSIFICATION

Hazardous areas with respect to gasses (e.g. in the vicinity of gas and liquid fuel lines and equipment) shall be classified according to IEC60079 API RP 505 or NFPA 497 and equipment installed within the classified zones shall be certified accordingly.

Induction motors

- Zone 1 (IEC); Class I, Zone 1 (API):
Explosion-proof type (EEx-d) or Increased Safety (EEx-e) or Pressurized Safety type (EEx-p).
- Zone 2 (IEC); Class I, Zone 2 (API):
- Explosion-proof type (EEx-d) or Increased Safety (EEx-e) or Pressurized Safety type (EEx-p).
Temperature Class – T3

• Lighting fixture

- Zone 1 (IEC); Class I, Zone 1 (API):
Explosion-proof type (EEx-d) or Explosion-proof + Increased Safety (EEx-de) or Increased Safety (EEx-e)
- Zone 2 (IEC); Class I, Zone 2 (API):
Explosion-proof type (EEx-d) or Explosion-proof + Increased Safety (EEx-de) or Increased Safety (EEx-e)
Temperature Class – T4

• Plugs and Sockets

- Zone 1 (IEC); Class I, Zone 1 (API):
Explosion-proof type (EEx-d) or Explosion-proof + Increased Safety (EEx-de)
- Zone 2 (IEC); Class I, Zone 2 (API):
As per Zone 1
Temperature Class – T4